

Segment 6 — Formal Validation: LOSO, Metrics & Bland-Altman

Spandan: Contactless Vital Sign Monitoring — EEE 312 DSP Project, BUET

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Why This Segment Exists

Segments 2-5 built the whole measurement chain: face detection, signal filtering, heart rate extraction, and an SpO2 calibration attempt. Every one of those segments produced a real number for a real subject and compared it, informally, against that subject’s own ground truth. What none of them did — deliberately, by design — was ask the harder, more honest question: **across every subject available, taken together, how good is this method really, and how do we know?** That is Segment 6’s entire job. It touches no video, no filter, no FFT — it consumes the numbers Segments 4-5 already produced and puts them through the same statistical machinery a real clinical device validation study would use.

Intuition First

Intuition first. Segments 2-5 were like a student solving practice problems one at a time and checking each answer against the back of the book. Segment 6 is the final exam: pool every problem attempted so far, grade all of them together, and report a real, defensible score — not “I got this one right,” but “here is my accuracy, my error spread, and my correlation with the answer key, computed honestly, with no problem’s answer key ever leaking into how that same problem was solved.”

What Bland-Altman Actually Shows, Versus a Simple Scatter

A plain scatter plot of predicted vs. true values answers one question: *do these two move together?* Pearson correlation is the number version of that same question. Both can be fooled by a **constant**

bias — two methods that always disagree by the same fixed amount will still correlate close to perfectly, because “moving together” says nothing about “being equal.”

A Bland-Altman plot asks a different, more clinically useful question: *how much do these two methods actually disagree, in absolute terms, and does that disagreement change depending on the value being measured?* It does this with a specific re-plotting trick:

- **x-axis: the mean of the two measurements**, $(\text{predicted} + \text{groundTruth}) / 2$ — a stand-in for “the best available estimate of the true value,” since a real ground-truth device (a pulse oximeter, a contact PPG sensor) is itself not perfectly noise-free.
- **y-axis: the difference between the two measurements**, $\text{predicted} - \text{groundTruth}$ — the actual disagreement, in the original units (bpm or SpO2%), not hidden inside a correlation coefficient.

Three horizontal reference lines complete the picture:

- **Bias** (the mean of all the differences) — is there a systematic tendency to over- or under-estimate?
- **Upper and lower limits of agreement** ($\text{bias} \pm 1.96 \times \text{SD of the differences}$) — the range within which about 95% of individual disagreements are expected to fall, under the standard assumption that the differences are roughly normally distributed.

Common Beginner Mistake

Common beginner mistake. Looking only at the correlation coefficient and concluding “high r means the method is accurate.” A method can be consistently 5 bpm too high on every single subject and still produce an r very close to 1.0, because r only measures whether the two variables rise and fall together — it is blind to a constant offset. Segment 6’s own HR results (see the Line-by-Line doc) show exactly this: CHROM/POS land an r of 0.995 on the pooled VIPL subjects while still carrying a real ~4 bpm average error. Bland-Altman’s bias line is what catches that a bare correlation number would hide.

Pause and Check Yourself

Pause and check yourself. If a method’s Bland-Altman plot showed a bias line sitting exactly at zero, but very wide limits of agreement (say, ± 20 percentage points for SpO2), would you call that method accurate? What does “zero average bias” actually guarantee, and what does it *not* guarantee, about any single individual prediction?

Why Pooling UBFC and VIPL for SpO2 Calibration Is the Right Move

`spo2/calibrateSpO2.m` fits a straight line, $\text{SpO2} = A - B \cdot R$, from whatever $(R, \text{SpO2})$ pairs it is handed. A linear fit’s reliability depends overwhelmingly on how many independent data points it has to learn from — 5 points (Segment 5’s DATASET_1-only attempt) is barely enough to define a line at all, and Segment 5’s own results showed exactly this instability: the fitted slope’s sign flipped depending on which single subject was held out.

The natural instinct is to worry that UBFC and VIPL are different enough (different cameras, different lighting, different populations) that mixing them “contaminates” the calibration. This instinct gets the tradeoff backwards for a **linear ratio-of-ratios calibration specifically**: what

kills this kind of fit is not moderate cross-dataset heterogeneity — it is having too few independent subjects to average that heterogeneity out over. More independent subjects, even somewhat different ones, is exactly what turns an unstable, sign-flipping 5-subject line into a more stable 8+ subject line. This is not a hand-wave — Segment 6’s own pooled result (1.23 percentage points pooled MAE, physiologically-correct-direction Pearson r of -0.755) is measurably tighter than Segment 5’s DATASET_1-only result (1.97 points, unstable slope sign), and the only thing that changed between the two is 3 more pooled subjects.

Intuition First

Intuition first. Estimating the average height of a population from 5 people, all from one city, gives you a shaky number that could easily be thrown off by one unusually tall or short person in that small sample. Estimating it from 8 people across two different cities gives you a more stable number, even though the two cities aren’t identical — because the extra independent data points do more to cancel out random sampling noise than the cities’ real differences do to bias the average. The same arithmetic applies to fitting a line through noisy points: more independent points beats a smaller, purer sample, up to the point where the *sources* themselves are so different they represent genuinely different physical relationships (not the case here — both datasets are the same red/blue ratio-of-ratios measured on human facial skin under ordinary lighting).

This does not mean dataset differences are irrelevant. Segment 6 pools UBFC and VIPL for the *fit itself* (a leave-one-out fold’s training set is genuinely drawn from both datasets together) precisely because a linear model with only two free parameters (A, B) cannot afford to throw away independent training subjects — but Segment 6 also reports **per-dataset breakdowns** (`results/metrics/segment6_spo2_loso_metrics.csv`’s UBFC and VIPL scope rows) specifically so the team can check whether accuracy holds up consistently across both datasets or quietly diverges on one of them. Pooling for training and still reporting per-source accuracy separately are not in tension — they answer two different, both useful, questions.

Pause and Check Yourself

Pause and check yourself. Segment 6’s per-dataset SpO2 breakdown shows UBFC’s MAE (1.22 points) and VIPL’s MAE (1.24 points) landing very close to each other. If instead one dataset’s MAE had come out dramatically worse than the other’s (say, 1.2 points for UBFC vs. 6 points for VIPL), would that be a reason to *stop* pooling them for the fit, or would it instead point toward a different, more specific problem to investigate first? What would you check before concluding “these datasets shouldn’t be pooled”?

The Calibrated-vs-Uncalibrated Distinction

This project’s two vital signs sit on genuinely different footing, and Segment 6’s evaluation design reflects that honestly rather than papering over it:

- **Heart rate is uncalibrated** — `heartrate/fftHeartRate.m` reads a frequency peak directly off a spectrum. Nothing about it is fit to any subject’s ground truth, ever. There is no parameter to leak, so HR is pooled across every available subject and scored directly with `validation/computeMetrics.m` — no held-out folds needed or appropriate.
- **SpO2 is calibrated** — `spo2/calibrateSpO2.m`’s A and B are learned from ground-truth data. Evaluating a calibrated model without holding subjects out is a well-known way to

produce artificially optimistic numbers, because the model can partially “memorize” the very subjects it’s being tested against. `validation/runLOSO.m` exists specifically to prevent this: every fold refits fresh on every other pooled subject before ever touching the held-out one.

Reporting both with the *same* MAE/RMSE/Pearson-r/Bland-Altman toolkit, while being explicit that they were produced by two different validation procedures for a real, physically-grounded reason, is more honest than either (a) applying LOSO to HR unnecessarily (padding out a report with folds that don’t change anything, since there is nothing to leak), or (b) skipping LOSO for SpO2 to save effort (silently accepting leaked, overly optimistic numbers).

How This Maps Onto Dr. Hasan’s Live Defense-Day Comparison

On defense day, the plan is a live, in-person comparison: Dr. Hasan’s own pulse oximeter against this project’s camera-based app, read in real time, side by side. It is worth recognizing explicitly what that live comparison actually *is*, statistically: it is a Bland-Altman-style comparison with $N = 1$.

One subject (Dr. Hasan himself, or whoever volunteers), one pair of readings (the app’s estimate, the oximeter’s reading) at one moment in time, produces exactly one point: one `mean` value, one `diff` value. That single point can be plotted on the exact same axes as every point in `results/figures/segment6_spo2_bland_altman_pooled.png`, and it will either fall inside this project’s own measured limits of agreement or it won’t.

Pause and Check Yourself

Pause and check yourself. If the live defense-day reading falls *outside* the pooled limits of agreement measured in Segment 6 (an unlucky but entirely possible outcome with a system trained on only 8 subjects), does that automatically mean the method is broken? What would $N=1$ be capable of proving or disproving on its own, versus what the pooled 8-point Bland-Altman study is capable of? (Hint: re-read what “95% of individual differences are expected to fall inside these two lines” actually implies about the other roughly 5%, and what a single sample can and cannot tell you about a distribution.)

This is exactly why Segment 6’s numbers, and this document’s own repeated insistence on stating N honestly, matter for the defense specifically: a single live comparison is real evidence, but it is statistically the thinnest possible amount of it — one draw from the same distribution Segment 6’s 8-subject (SpO2) or 3-subject (HR, today) pooled study is already trying to characterize. Walking in with the pooled Bland-Altman plot and limits of agreement already in hand means a single live reading — good or bad — can be explained and contextualized on the spot, rather than treated as the entire verdict on the project.